

DIVAR CASTRO RODAS

PhD(c) in Engineering (Georesources) · M.Sc. in Geographic Information Systems · Environmental Ocean Engineer

CONTACT

divecast@espol.edu.ec
Guayaquil, Ecuador
LinkedIn: /in/divar-castro-rodas
ORCID: 0000-0002-2197-2277
Lattes (CNPq)
ResearchGate
GitHub: /DivarCastro

LANGUAGES

Spanish — Native
English — Fluent (B2)
Portuguese — Intermediate

TECHNICAL SKILLS

- GIS: QGIS, ArcGIS Pro
- Programming: Python, R
- Google Earth Engine
- PostgreSQL / PostGIS
- Remote sensing and satellite imagery
- Certified drone (UAV) pilot
- Advanced office software

SOFT SKILLS

- Teamwork
- Analytical skills
- Effective communication
- Proactivity
- Scientific coordination

OTHER INTERESTS

- Environment
- Beach clean-ups
- Technical visits

PROFESSIONAL PROFILE

PhD candidate in Engineering, Georesources track (ESPOL). Oceanographic and meteorological information analyst with 9 years of experience in Geographic Information Systems and remote sensing applied to environmental and coastal monitoring. Experience in scientific coordination, statistical analysis, management of funded research projects, and university teaching. Active collaborator in geospatial research projects in Ecuador and Brazil.

WORK EXPERIENCE

Research Technician 2 — ESPOL

Guayaquil, Ecuador | 2024 – Present

- Authoring indexed publications on GIS and remote sensing.
- Managing funding proposals for research projects.
- Leading the development of the Ecuadorian Geoportal of Ocean-Atmospheric Parameters (GEPOA) for satellite monitoring of ocean-atmospheric variables in Marine Protected Areas.
- Collaborator in the Brazilian HIDRA project (call CNPq/MCTI/FNDCT No. 06/2026, Faixa B) on ecosystem water use efficiency (Random Forest, XGBoost, SHAP, Google Earth Engine).
- Collaborator in the STARS project (Spatial Tools for Assessing Risks in Sudden Natural Hazards), funded by Erasmus+ (Capacity Building) and led by the University of Lisbon, developing GIS and BIM educational modules for risk assessment of sudden natural hazards (earthquakes, floods, landslides) at universities in Ecuador and Colombia.
- Participation in community outreach projects.

Lecturer — Tecnológico Universitario Espíritu Santo

Guayaquil, Ecuador | 2022 – Present

- Creating syllabi and content for the courses taught.
- Academic planning and student follow-up.
- Organizing extracurricular activities.
- Community outreach projects.

Oceanography and Meteorology Analyst — INOCAR

Guayaquil, Ecuador | 2022 – 2024

- Using GIS tools and programming for navigation aids and vessel drift scenario modeling.
- Maintenance of Ecuador's tide gauge and buoy network.
- Database updating and collection of in situ and satellite data.
- Participation in regional and coastal oceanographic cruises.
- In charge of the directorate's spatial object catalog.

Bioassay Coordinator & Scientist — CENAIM-ESPOL

Santa Elena, Ecuador (Internship) | 2017 – 2022

- Statistical analysis and management of the center's spatial information.
- Technician in charge of bioassays and the oceanographic/meteorological station.
- Technical follow-up of the Annual Operating Plan.

EDUCATION

PhD in Engineering, Georesources track — ESPOL

2024 – Present

- Thesis: "Identification and Prediction of Urban Growth in Greater Guayaquil (IPUS)"

Diploma in AI Applied to Teaching — Universidad de las Américas

2025

Environmental Ocean Engineer — Universität Salzburg & USFQ

2019 – 2021

M.Sc. in Geographic Information Systems and Science — ESPOL

2011 – 2015

PUBLICATIONS

Castro-Rodas, D., Pico Molineros, A., Thottolil, R., Bulatov, D., de Paula, E. M. S., & Velastegui-Montoya, A. (2026). *Comparison of Supervised Classification Algorithms for Land Use and Land Cover in Guayaquil: An Assessment with Landsat and MapBiomas*. Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLIX-B3-2026, 565–572. DOI

Castro-Rodas, D., Velastegui-Montoya, A., Guimarães, Y. C., Nascimento, E. S., Pereira, M. D. A., Santos, L. B. L., Alvarez, C., & Sanclemente Ordoñez, E. (2026). *Detecting Urban Change with Open-Source GIS Programs: A Case Study in Guayaquil*. ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci., X-3/W4-2025, 49–54. DOI

Peña-Villacreses, G., Velastegui-Montoya, A., Castro-Rodas, D., Freire-Granda, D., de Souza, I. N., & Pereira, L. C. C. (2026). *GIS Mapping of Allochthonous Marine Litter across the Brazilian Amazon Coast*. ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci., X-3/W4-2025, 279–284. DOI

Pineda, F. V. C., Loor, D. S. C., Añez, A. E. P., & Rodas, D. C. (2023). *Disruptive Technologies and Businesses: Toward an Accelerated Transformation of Society and the Market*. In Technology, Business, Innovation, and Entrepreneurship in Industry 4.0 (Guarda, Fernandes & Augusto, Eds.). Springer, Cham. DOI

Castro Loor, D., Calderón Pineda, F., Castro Rodas, D., & Rodas Idrovo, R. (2022). *Posicionamiento de emprendimientos en artesanos, provincia de Santa Elena*. Revista Ciencias Pedagógicas e Innovación, 10(2), 45–50. DOI

Abad, R., Jaramillo, K. B., Castro, D., Sánchez, J. A., & Rodríguez, J. (2022). *Octocoral Distribution Patterns at the Equatorial Front (Tropical Eastern Pacific): Muricea and Leptogorgia*. Oceans, 3, 218–230. DOI

Ormaza-González, F. I., Castro-Rodas, D., & Statham, P. J. (2021). *COVID-19 Impacts on Beaches and Coastal Water Pollution at Selected Sites in Ecuador, and Management Proposals Post-Pandemic*. Frontiers in Marine Science, 8, 710. DOI

TALKS AND EVENT PARTICIPATION

IGARSS 2026 — Invited speaker (IEEE). Washington, D.C., USA, August 2026.

XXV ISPRS Congress — Invited speaker (ISPRS). Toronto, Canada, July 2026.

Latin America GRSS and ISPRS Remote Sensing Conference (LAGIRS) — Tutor and invited speaker. Foz do Iguaçu, Brazil, November 2025.

Workshop: Downloading and processing satellite data with Python and GEE — Invited trainer, FICT-ESPOL. Guayaquil, Ecuador, October 2025.

The 2nd International Electronic Conference on Land — "A Comparison of Supervised Classification Algorithms in Guayaquil Land Use and Land Cover Data" — Invited speaker, MDPI. Online, Switzerland, September 2025.

AmeriGEO Week 2025 — "Building and using the RuPO Hub to address marine plastic" — Invited speaker, GEO. Bogotá, Colombia, August 2025.

Pre-Antarctic Course — "Teleconnections with Antarctica" — Invited speaker, INOCAR. Guayaquil, Ecuador, December 2023.

Basic GIS course with QGIS — Invited trainer, UPSE. Santa Elena, Ecuador, December 2023.

World Oceans Day — "Oceanography in Ecuador: what are we doing as a country?" — Invited speaker, ESPOL. Guayaquil, Ecuador, June 2023.

Ecosystem Services — "Remote Sensing and Valuation of Ecosystem Services" — Invited speaker, CODSpace-UAndes. Online, Colombia, September 2020.

COURSES AND TRAINING

- Certified drone pilot (40 h) — DGAC
- Drone operation and handling (8 h) — ESPOL
- Programming Language and Artificial Intelligence (8 h) — ESPOL
- Oceanography Symposium 2024 (16 h) — ESPOL
- Oceanography of the Equatorial Region (20 h) — ESPOL
- Oceanographic Information Analysis and Processing with Python (40 h) — ESPOL
- Design and implementation of a numerical model of wave transformation and overtopping on the Ecuadorian coast — SWASH model (80 h) — INOCAR
- NASA ARSET: Introduction to NASA Earth Observations and Tools for Operational Wildfire Monitoring and Management
- NASA ARSET: Monitoring Global Terrestrial Surface Water Height using Remote Sensing
- NASA ARSET: Using Google Earth Engine for Land Monitoring Applications
- NASA ARSET: Remote Sensing of Coastal Ecosystems
- ESRI: 3D Visualization, Animation, Labeling, Layer Management, Raster Data and Coordinates in ArcGIS Pro

AWARDS AND FELLOWSHIPS

- Doctoral fellowship in Brazil, starting August.
- Best oral presentation — The 2nd International Electronic Conference on Land (MDPI Land, Switzerland).
- Humboldt-Caldas Medal for the best work in Biogeography of Ecuador and Colombia (Colombian Academy of Exact, Physical and Natural Sciences, Bogotá).
- Honor diploma for the best graduate of the program, 2016–2017 (ESPOL).

REFERENCES

Jenny Rodríguez, Ph.D. — Research Professor, CENAIM-ESPOL | jenrodri@espol.edu.ec

Mercy Borbor Córdova, Ph.D. — Research Professor, FIMCM-ESPOL | meborbor@espol.edu.ec

Andrés Velástegui Montoya, Ph.D. — Research Professor, FICT-ESPOL | dvelaste@espol.edu.ec